

CS63 Spring 2026

Lab 8: Convolutional Neural Networks

Jaden Jones and Reishiro Kawakami

1 Data Set

The dataset contains 70,000 clothes images of size 28 x 28 pixels, with each image labeled with a target label. The possible target labels are as follows:

- 0 = t-shirt
- 1 = trouser
- 2 = pullover
- 3 = dress
- 4 = coat
- 5 = sandal
- 6 = shirt
- 7 = sneaker
- 8 = bag
- 9 = ankle boot

2 Network

The best performing network we created was a result of pooling the "shirt" and "t-shirt" categories into, which seemed vague from a human perspective, into the more general "pullover" category. This was partly motivated by the data in the confusion matrix, which indicated that these two categories were getting easily confused for each other. Additional accuracy was gained by increasing the number of training epochs from 5 to 15.

Layer (type)	Output Shape	Param #
input (InputLayer)	(None , 28 , 28 , 1)	0
conv1 (Conv2D)	(None , 24 , 24 , 8)	208
pool1 (MaxPooling2D)	(None , 12 , 12 , 8)	0
flatten (Flatten)	(None , 1152)	0
hidden (Dense)	(None , 50)	57,650
output (Dense)	(None , 10)	510

Total params: [58,368](#) (228.00 KB)

Trainable params: [58,368](#) (228.00 KB)

Non-trainable params: [0](#) (0.00 B)

Figure 1: The summary of the network

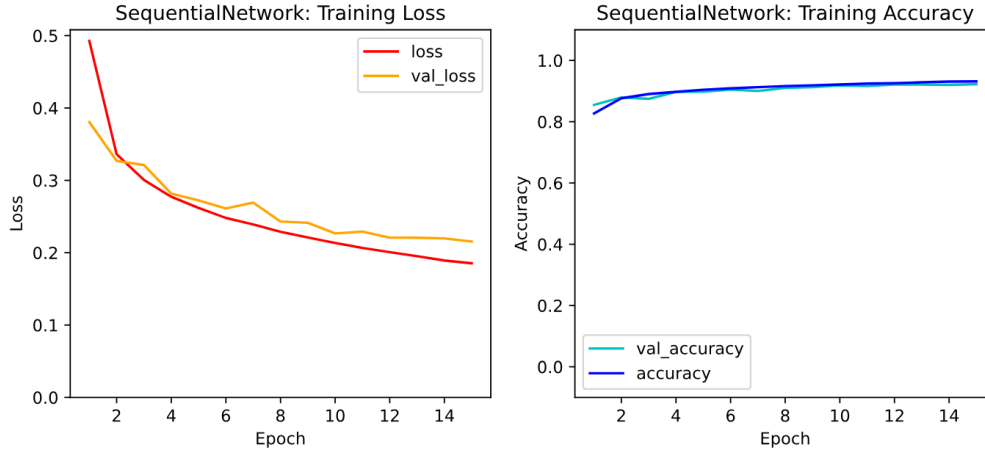
3 Training

Run	Accuracy	Val Accutacy
1	0.932	0.922
2	0.935	0.923
3	0.936	0.927
4	0.935	0.927
5	0.936	0.917
Average	0.935	0.923

4 Evaluation

The network performed well on trousers and sneakers. This is probably because these clothes have distinct feature maps. The trousers have the edges for legs, which is a distinct feature of trousers in the dataset, and the model was focused on the feature (see figure 4). It may have been more difficult if the dataset included shorts. The sneakers had high accuracy probably because of the relatively distinctions between boots (figure 5), sneakers (figure 6), and sandals (figure 7). The feature maps showed that the model was focusing on the upper edge of the shoes, which was consistenyl a steep curve for boots, a little blurry for sandals, and rather flat for sneakers.

On the other hand, the network performed poorly for coat, pullover, and dress in terms of



Epoch 15/15 accuracy: 0.9315500259399414 - loss: 0.1853506863117218 - val_accuracy: 0.9223999977111816 - val_loss: 0.21548894047737122

Figure 2: A typical learning graphs

the raw counts. This seems relatively reasonable considering their similarities – we felt that even humans would struggle to distinguish between these three for some of the images. When we look at the feature maps, we can tell that the model is focused on the arms (and a little bit of the collars) when it comes to coat and pullover. However, arms are common features between coat and pullover. The feature map for dress also focuses on the edges around the torso of the dress, which looks like the arms of pullovers. Therefore, we think the model is focusing on the parts (i.e. arms) that don't really let the model to distinguish between these categories.

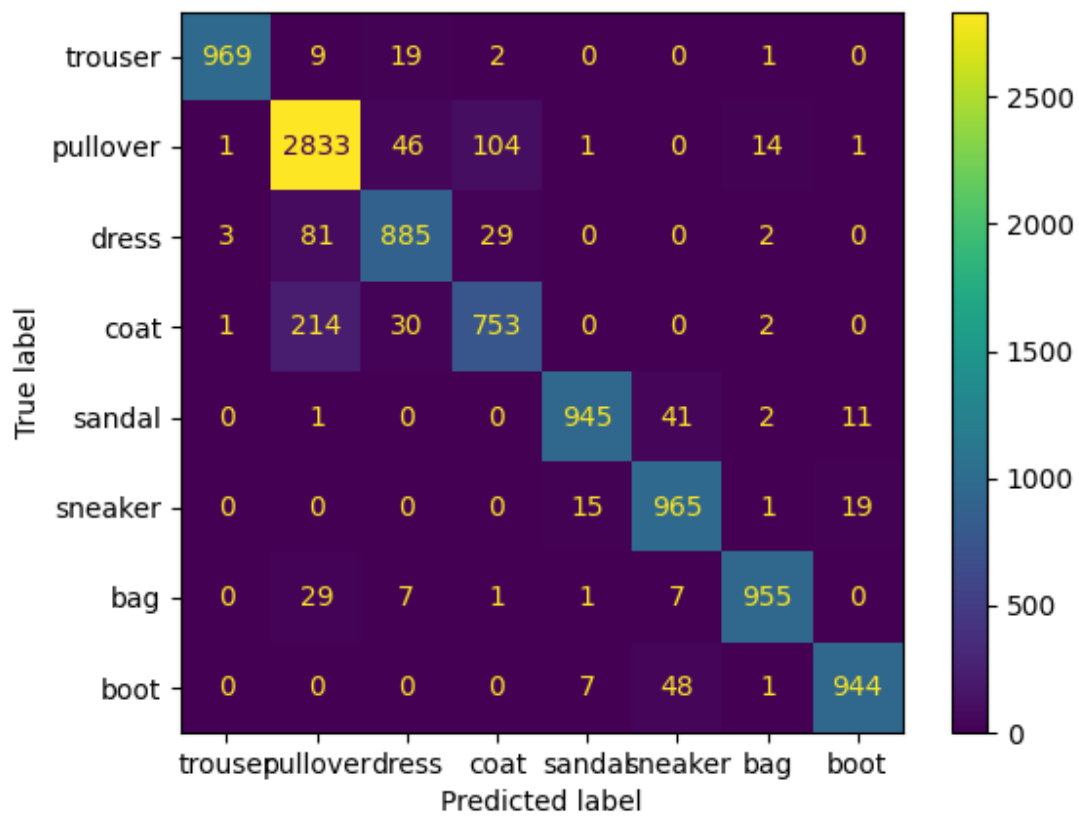


Figure 3: The confusion matrix

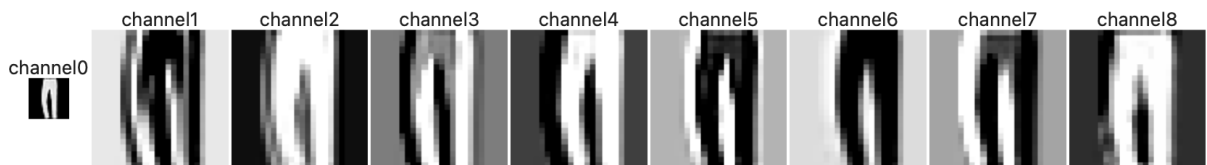


Figure 4: The feature map of a trouser

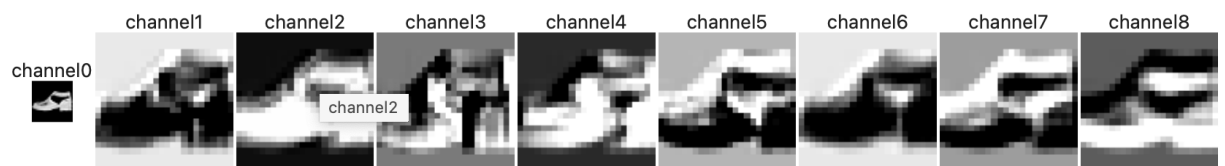


Figure 5: The feature map of a boot

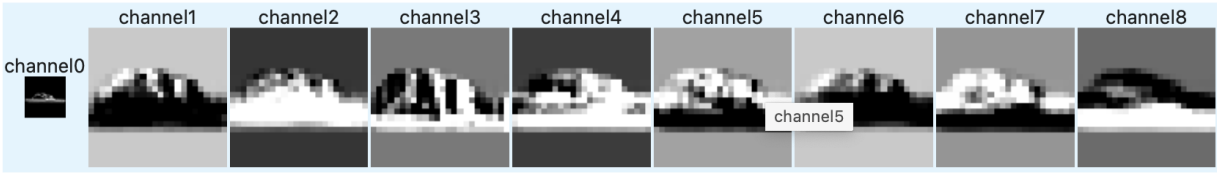


Figure 6: The feature map of a sandal

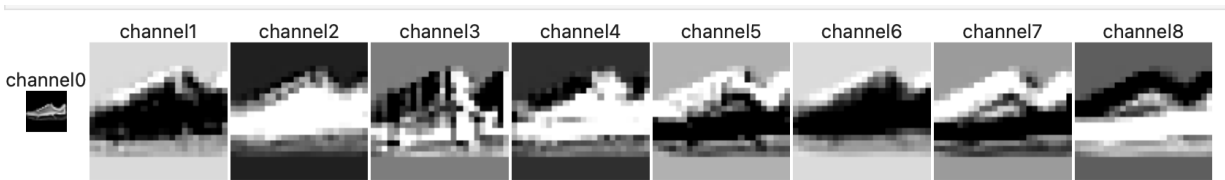


Figure 7: The feature map of a sneaker

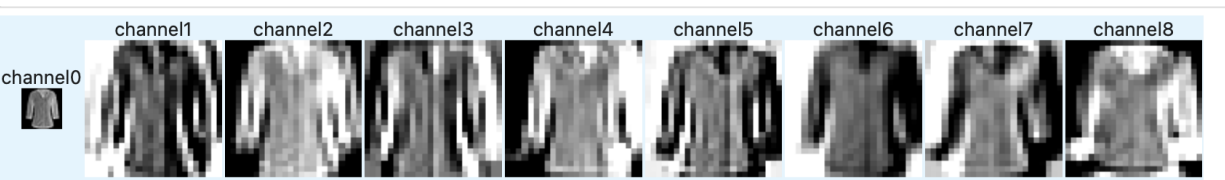


Figure 8: The feature map of a pullover

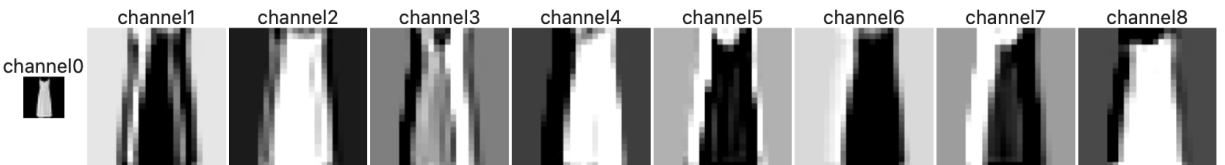


Figure 9: The feature map of a dress